

LOSSIEMOUTH, United Kingdom

WMO: 030680

Lat: 57.7114N

Lon: 3.3233W

Elev: 23

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.1	26.8	19.7	14.8	26.3	22.7	17.1	30.4	39.8	41.6	34.6	41.7	4.7	240	0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	71.9	61.6	68.8	60.0	66.3	58.6	63.2	69.5	61.5	66.9	59.9	64.6	13.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	78.8	66.1	59.0	74.5	64.0	57.5	70.6	62.3	28.7	69.6	27.5	67.2	26.3	64.8	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.9	27.2	24.3	DB	17.7	78.7	6.1	4.2	13.3	81.7	9.7	84.1	6.3	86.5	1.9	89.5
			WB	16.9	66.8	6.1	2.7	12.5	68.8	9.0	70.3	5.5	71.8	1.1	73.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	38.9	39.4	41.8	45.7	49.6	54.3	58.2	57.8	54.4	49.1	43.4	39.0
	DBStd	8.55	5.07	5.32	5.03	5.05	4.96	4.20	4.29	3.96	4.18	4.87	5.25	6.49
	HDD50	1754	344	299	257	148	67	7	0	1	8	75	205	343
	HDD65	6330	808	717	719	580	476	320	218	227	318	494	648	805
	CDD50	911	0	2	4	18	56	138	254	242	141	47	7	2
	CDD65	10	0	0	0	0	0	0	6	3	1	0	0	0
	CDH74	51	0	0	0	0	3	3	28	16	1	0	0	0
	CDH80	4	0	0	0	0	0	0	3	1	0	0	0	0
Wind	WSAvg	11.8	13.1	13.4	13.4	12.3	11.1	10.8	9.8	10.1	11.1	12.0	12.6	12.4
Precipitation	PrecAvg	29.1	2.2	2.0	1.9	1.9	2.1	2.6	2.6	2.8	2.8	3.2	2.6	2.3
	PrecMax	35.4	4.2	4.1	3.3	5.4	4.7	5.0	5.1	7.2	11.9	5.6	4.2	3.7
	PrecMin	21.8	0.5	0.8	0.6	0.7	0.6	1.2	1.0	0.7	1.0	0.9	0.7	0.6
	PrecStd	3.2	1.0	0.8	0.6	1.1	0.9	1.1	1.3	1.4	2.0	1.3	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	56.5	58.9	64.8	70.3	72.3	78.1	76.1	71.3	64.7	58.9	56.5
		MCWB	49.8	50.7	50.3	53.7	57.0	60.6	64.1	63.8	61.1	57.4	53.9	51.4
	2%	DB	51.5	53.7	55.0	60.3	64.7	67.8	72.1	70.8	67.1	61.4	55.7	53.9
		MCWB	47.5	49.0	48.5	51.7	55.6	58.8	62.2	61.7	59.2	55.1	51.4	49.7
	5%	DB	49.2	51.0	52.4	57.2	61.0	64.7	68.7	67.6	64.1	59.0	53.5	51.2
		MCWB	45.7	46.6	46.9	49.8	53.3	56.9	60.5	59.6	57.2	53.7	49.6	47.5
	10%	DB	46.8	48.0	50.3	54.4	57.9	62.0	65.9	64.8	61.7	56.8	51.5	48.6
		MCWB	43.7	44.2	45.4	48.2	51.5	55.3	59.1	58.2	55.6	52.2	48.2	45.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	51.7	51.7	55.1	59.7	63.1	66.3	65.2	62.5	58.1	54.5	52.0
		MCDB	53.8	55.5	57.4	62.5	66.1	69.5	74.6	73.0	69.4	63.1	58.2	55.7
	2%	WB	47.9	49.1	49.3	52.7	56.4	59.8	63.6	62.9	60.1	56.1	52.0	49.9
		MCDB	51.1	53.0	54.1	59.2	62.7	66.0	69.9	68.9	65.6	60.1	55.1	53.6
	5%	WB	45.9	46.9	47.5	50.5	54.3	57.9	61.8	61.1	58.2	54.5	50.1	47.7
		MCDB	48.8	50.8	51.8	56.0	59.9	63.2	67.1	66.2	62.6	58.0	52.8	50.6
	10%	WB	44.0	44.4	45.7	48.6	52.3	56.2	60.0	59.4	56.7	52.9	48.5	45.6
		MCDB	46.6	47.6	49.7	53.3	57.0	60.9	64.6	63.4	60.4	56.2	51.2	48.3
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.9	11.9	12.5	12.8	11.6	12.2	12.4	12.3	11.6	10.2	10.3
		MCDBR	11.5	13.0	15.1	17.3	18.5	17.1	17.6	17.0	16.0	14.2	12.3	12.2
		MCWBR	9.5	10.0	10.3	10.4	10.5	9.9	9.7	9.7	9.9	10.1	10.2	10.3
	5% WB	MCDBR	11.1	12.6	14.2	16.1	16.9	15.7	15.8	15.7	14.7	13.2	11.6	12.1
		MCWBR	9.5	10.2	10.4	10.3	10.4	9.8	9.5	9.7	10.1	10.2	10.0	10.4
Clear-Sky Solar Irradiance	taub	0.307	0.332	0.355	0.375	0.374	0.372	0.376	0.364	0.349	0.335	0.309	0.292	
	taud	2.250	2.346	2.345	2.325	2.373	2.408	2.405	2.447	2.489	2.446	2.298	2.180	
	Ebn at Noon	171	224	251	265	273	275	271	267	253	222	170	136	
	Edh at Noon	16	22	29	34	35	34	34	30	25	20	15	12	
All-Sky Solar Radiation	RadAvg	134	335	666	1097	1481	1455	1408	1111	773	408	174	91	
	RadStd	16	32	70	104	79	119	113	71	69	34	14	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page